CODEALPHA

ROBOTICS & AUTOMATION INTERNSHIP

TASK 2 — PROJECT REPORT

3-DOF Robotic Arm Simulation for Pick-and-Place

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Abstract

This project demonstrates a simple three-degree-of-freedom (3-DOF) robotic arm performing a pick-and-place operation in a browser-based simulation. The arm is represented as three connected links with rotary joints and a gripper at the end. The sequence moves to a pickup location, picks up an object, carries it to a target zone, releases it, and returns to a home pose. This is an educational visualization and does not control physical machinery.

1. Introduction

Robotic arms are used in manufacturing for material handling, assembly, packaging, inspection, and machine tending. Pick-and-place is a basic automation task in which an object is collected from one location and moved to another. Simulation helps students understand joint motion and end-effector movement without needing a physical robot.

2. Objectives

The objectives are to model a simple 3-DOF robotic arm, identify its joints and links, demonstrate a pick-and-place sequence, observe how joint-angle changes affect the arm pose, and document the result with screenshots.

3. Software and Model

The project uses a self-contained HTML simulation that runs in a modern browser such as Chrome or Edge. No CAD package is required. The model contains three links, three revolute joints, a base, a gripper, a pickup object, and a target zone. Link lengths are simulation units, not real-world millimetres.

4. Degrees of Freedom (DOF)

A degree of freedom is an independent motion variable that can be controlled. In this planar model, Joint 1 is shoulder rotation, Joint 2 is elbow rotation, and Joint 3 is wrist rotation. Each is a revolute joint. The gripper opens and closes as an end-effector action and is not counted as one of the three arm joints.

5. Working Principle

The arm is represented as connected links. Changing a joint angle changes link orientation and end-effector position. The demonstration moves to a pickup pose, visually attaches the object to the gripper, moves to the placement pose, releases the object in the target zone, and returns home. It is simplified and does not model realistic forces, friction, or industrial safety functions.

6. Procedure

1. Save the supplied HTML file. 2. Open it in Chrome or Edge. 3. Click Run Pick-and-Place. 4. Observe the arm move to the object. 5. Observe the object travel with the gripper. 6. Observe release and return to home. 7. Click Save Screenshot (PNG) to save the canvas. 8. Capture additional screenshots at the pickup and placement stages.

7. Expected Results and Observations

The simulation should visually demonstrate a complete pick-and-place cycle. The arm changes pose through three rotary joints, the object is shown at the gripper during transport, and the object is released at the target zone. Add screenshots after running the simulation. Do not describe this as a physical robot test; the result is browser-based visualization.

8. Applications

Similar operations are used in packaging lines, sorting, warehouse handling, assembly stations, machine tending, and transferring components between conveyor belts. Real applications require appropriate grippers, sensors, motion controllers, risk assessment, and safety systems.

9. Limitations and Future Improvements

This is a simplified 2D educational model. It uses predefined poses and does not represent a fully validated 3D industrial robot, detailed inverse kinematics, realistic collision detection, payload limits, or certified safety controls. Future improvements could include a 3D model, sensor-triggered pickup, adjustable object positions, collision checks, and more advanced motion planning.

10. Conclusion

The project demonstrates a robotic arm with three rotary degrees of freedom and a gripper. It shows the main stages of a pick-and-place operation and helps explain how joint movements combine to position an end-effector. It provides a foundation for further work with sensors, controllers, and 3D robot models.

References

1. CodeAlpha, “Robotics & Automation Tasks & Instructions,” internship task sheet. 2. International Federation of Robotics, “Industrial Robots,” https://ifr.org/industrial-robots. 3. ISO, ISO 10218 series, safety requirements for industrial robots and robot applications, https://www.iso.org.

Screenshot Checklist

- Screenshot 1: Simulator opened at home pose.
- Screenshot 2: Object picked / transport stage.
- Screenshot 3: Object placed and completion status visible.